Ravan.ai (Agni) vs Vistrow Voice — platform audit

Date: 24 October 2026. Author: Claude (runtime session), for cross-verification by Codex.

How to read the evidence marks

• **[S]** seen on a Ravan screen, logged in to our own Prophunt LLP account (Free plan, 0 credits), 24 Oct 2026. Nothing was bought, called, scraped or changed except switching our own test agent's model to read voice lists, then restoring Premium + Iris.
• **[J]** read from the model-catalog JSON the Ravan dashboard loads for our session (`/api/proxy/available-llm-models/with-voices`).
• **[D]** Ravan public docs (docs.ravan.ai, including llms.txt and openapi.json).
• **[A]** Ravan's in-app "Ask Agni" assistant. It gave 2 wrong answers out of 7 (reminder message, IVR direction), so it is used only where [S], [J] and [D] are silent.
• **[V]** verified by reading our code on `origin/main` (commit 9ef663cf) or production Railway variables on 24 Oct 2026.
• **[I]** inference by the author; not in any source. Codex: please challenge these.
Codex cross-check: every [V] row names a file; run `git show origin/main:<file>` and confirm. Every [J] row can be re-read by opening app.ravan.ai while logged in and viewing the request in DevTools → Network → `with-voices`.

1. Ravan's vendors (the key finding)

Every "Agni" model is a renamed third-party model. [J]: each model carries `realLlmModel`; each voice carries `realVoiceId` / `realVoiceName`. Catalog fields also include `sessionMode`, `sttModel`, `pipelineLlmModel`, `ttsModel`, `realtimeModel`, `showInUi` (the per-stage fields were empty for every model). Re-read 24 Oct after Codex's check: 11 entries, 9 with `showInUi=true`; `sessionMode` is `pipeline` for 8 and empty for `sarvam_pipeline`, `murf` and `Soniox`. The catalog shows vendor aliases, not the hidden execution pipeline.

Agni model	Credits/min (Rs at Rs10/credit)	realLlmModel [J]	Real voices seen [J]	Author's reading [I]
Ultra	1 (Rs10)	`gemini` — description: "Live API … 30 HD voices across 24+ languages … thinking, proactive audio, affective dialog, function calling"	Aoede, Laomedeia, Kore, Fenrir, Leda, Orus, Callirrhoe, Zephyr, Puck, Charon (27)	Google Gemini Live API, newest version
Premium (marked RECOMMENDED)	1 (Rs10)	`gemini_3.1`	same Gemini voice set (27)	Gemini 3.1 Flash Live
Premium Lite	0.7 (Rs7)	`gemini_2`	same Gemini set (27)	Gemini 2.x Live
Duplex	1 (Rs10)	`gpt-live-1`	ripple, quartz, Ash, tempo, Nira, Nuri (16)	OpenAI GPT-Live 1 (listed on OpenAI's pricing page at $0.05/min)
5.0 — description: "It is the Cheapest."	0.6 (Rs6)	`boson`	hashed ids `voice_d0753a4e`… plus Yashika, Rupali, Arjun, Ram Gujarati, Riya Hinglish, Riya Marathi, Ram Bengali, Harsh Hinglish 1/2 (14, all "No preview")	Boson AI (Higgs Audio) with custom cloned Indian voices
5.0 Lite	0.5 (Rs5)	`svara_pipeline`	`sv_bnaqa3fy`, `sv_enhdbrj5`, Sipoto, Arjun (9)	Svara by name; Kenpath's model specifically is unproven (Codex: `sv_` ids do not match Kenpath's published voice names)
Lite 4.0	0.55 (Rs5.50)	`sarvam_pipeline`	kavya, amit, rahul, priya, ritu (8)	Sarvam Bulbul — the same vendor and voices as our "Bharat"
Lite 3.0 (beta) — "Budget friendly"	0.45 (Rs4.50)	`murf`	de-DE-björn, pa-IN-harman, fr-FR-justine … (118)	Murf AI
Lite 2.0 — "Fast, low-cost multilingual voice model"	0.45 (Rs4.50)	`Soniox`	Elise, Maya, Nina, Claire, Mina, Mateo, Meera (28)	Soniox (sells STT and TTS, so one vendor can cover both; the LLM is not shown)
4.0 (hidden, showInUi=false)	1	`gemini-3.1-flash-lite`	Leda, Laomedeia, Rasalgethi (14)	Gemini 3.1 Flash-Lite
3.0 (hidden, showInUi=false)	1	`gemini_pipeline`	Zephyr, Alnilam, Algenib (27)	STT → Gemini → Gemini TTS chain

Supporting observations:

• [S] Switching the test agent to Lite 2.0 removed the Emotion and Accent controls and changed the voice to Elise; Ultra, 5.0 and Duplex kept them. 5.0 is `boson`, so Emotion is not Gemini-only; how each model implements it is not known.
• [D] Agents have both `model` and `s2sModel` fields; example `gpt-4o` / `gpt-4o-realtime`. Post-call analysis runs on `gpt-4o-mini` [S][D].
• [D] Customers point their SIP carrier at `sip:…sip.livekit.cloud`; the create-call API returns a LiveKit access token and room id; the error-codes page explains LiveKit error wrappers. Platform = LiveKit Cloud.
• [S] Every dashboard API call is proxied through `app.ravan.ai/api/proxy/…`, so vendor hosts are not visible from normal browsing.
• [S] Website tracking and media: `link.quickadpro.com`, `backend.leadconnectorhq.com`, `assets.cdn.filesafe.space` = GoHighLevel/LeadConnector. Voice previews are static MP3 files on that CDN.
• [D] Trust Center sub-processor list names only AWS, Twilio, OpenAI, Stripe — it omits LiveKit, Google, Boson, Sarvam, Murf, Soniox, Svara.
• [J] The catalog aliases and voice ids strongly indicate the vendors above. They do not rule out Ravan's own routing, prompts, fine-tuning or processing, so the "in-house engine" and "<300 ms" claims are unproven either way.
Consequences for Vistrow:

• Their flagship (Premium/Ultra, Rs10/min) is the model we sell as Vistrow Live (Gemini Live). We run Gemini Live 3.8; their recommended model is 3.1. Our measured cost on Live 3.8 is Rs4.7–5.3/min (call 1418 and later tests).
• Their Indian-voice tier (Lite 4.0) is Sarvam, which we already run as Bharat.
• Vendors we do not have: Boson/Higgs Audio (cloned Indian voices), Murf, Soniox, Svara.
• Catalog metadata strongly indicates Ravan depends on the same kind of third-party providers we do; it does not prove they have no in-house component.

1b. Vistrow's own models and voices → vendor (from code on origin/main, 24 Oct 2026)

Our customer-facing names are vendor-neutral (owner rule 15 Oct). This table is the decoder. Codex: verify ids in web-demo/src/lib/agentOptions.ts (MODEL_OPTIONS, ADMIN_ONLY_MODELS, RETIRED_MODELS), server/voice_catalog.py (prefixes, tiers, _OWNER_ONLY_PREFIXES) and agent/main.py (LLM/STT/TTS builders around lines 2060-2250 and 2447-2495).

Vistrow name (what customers see)	Internal id / prefix	Real vendor + model	Who can use it
Vistrow Bharat (LLM)	`sarvam/sarvam-105b-conversations`	Sarvam 105B through the OpenAI-compatible plugin; falls back to OpenAI `gpt-5.4-nano`	All customers; the only LLM in the tenant picker
GPT-5.4 nano / mini	`gpt-5.4-nano`, `gpt-5.4-mini`	OpenAI	Kept for existing agents; nano is the fallback LLM
Vistrow Nano Lab	`gpt-5-nano`	OpenAI	Owner-only (ADMIN_ONLY_MODELS)
Gemini 3.5 Flash-Lite / 3.6 Flash / 3.8 Flash	`gemini-3.5-flash-lite`, `gemini-3.6-flash`, `gemini-3.8-flash`	Google Gemini API (`google.LLM`)	Combo tests only; not in the tenant picker
Vistrow Live 3.8	`gemini-live:gemini-3.8-live`	Google Gemini Live API, speech-to-speech `RealtimeModel`	Dashboard picker: owner-only (agentOptions.ts ADMIN_ONLY_MODELS, lines 323-332). Server: any tenant may save it (token_api.py `_TENANT_REALTIME_MODELS`, owner note 19 Oct "open to every plan"). Billed 2 credits/min (agent/plan_policy.py). Same vendor as Ravan Ultra/Premium
Vistrow Live 2.5 / 3.1	`gemini-live`, `gemini-live:gemini-3.1-flash-live-preview`	Google Gemini Live	Retired 19 Oct
Groq	`groq/*`	Groq	Owner-only; removed from the picker 7 Sep
Speech recognition, default	`stt_provider = "sarvam"`	Sarvam saaras v3, 17 Indian codes	All
Speech recognition, alternative	`stt_provider = "google-chirp3"`	Google Cloud Speech Chirp 3	All, per agent
Bharat voices (tab Bharat)	stored v3 short names (`ritu`, `priya`, `rohan`) and `sarvam4:<name>`	Sarvam Bulbul **v4 Flash** for both: short names map through `voice_catalog.SARVAM_V3_TO_V4` (agent/main.py ~3290). Ravan Lite 4.0 uses the same short names	All; tier standard
Vistrow HD	`google:chirp3:<Persona>` (Callirrhoe, Sana, Aoede)	Google Cloud Text-to-Speech Chirp 3 HD, service account	All; tier standard

Vistrow name (what customers see)	Internal id / prefix	Real vendor + model	Who can use it
Native voices	`google:<lang>-IN-Standard-X` (Anika, Kabir)	Google Cloud Text-to-Speech Standard	All; tier lite
Mira / Arin Expressive	`google:kore`, `google:charon`	Google Gemini 2.5 Flash TTS (voice_catalog `GEMINI_TTS_MODELS`)	Tier premium + `preview` → owner-only for new selection (token_api `_guard_voice_tier`); saved agents keep it
Studio Lite	`google38:<voice>` (for example `en-in-csagent-3` = Anjali)	Google Gemini 3.8 Flash-Lite TTS through the Gemini Interactions API (`GEMINI_API_KEY`)	Owner-only (preview) until Gemini API Tier 2
Studio	`google38flash:<voice>`	Google Gemini 3.8 Flash TTS through the Interactions API	Owner-only (preview); tier premium 2x
Mira / Arin Next (Preview)	`google31:kore`, `google31:charon`	Google `gemini-3.1-flash-tts-preview` (agent/main.py `_GOOGLE_31_MODEL`)	`preview` → owner-only
Premium (tab Premium)	`elevenlabs:<voice id>` (Tara, Riya, Saavi, Bunty)	ElevenLabs	Owner-only for new selection; saved agents keep theirs. Premium tier also holds Google preview voices; customers can pick zero premium voices (bug 3)
Cloned voices	custom_voices table	Sarvam voice cloning; first clone free on Growth/Scale, later clones 25 credits each (server/custom_voices.py:70-73)	Off for customers unless `VOICE_CLONE_OPEN` is set (read unset on Railway 24 Oct; Codex could not check production)
Call platform	LiveKit Cloud project Artha Voice; agents CA_TGdpVhSdxDyS (phone and tenant) and CA_53d8HgBktjZ7 (website demo)	LiveKit, Ship plan $50/mo	—
Telephony	EnableX SIP trunk into LiveKit SIP	EnableX, Rs0.25/min each way	Customers bring their own number
Post-call analysis	`gpt-5.4-mini` (server/call_intelligence.py); KB Q&A drafting `gpt-5.4-nano`	OpenAI	—

Same-vendor pairs: Agni Ultra / Premium / Premium Lite = Vistrow Live (Gemini Live). Agni Lite 4.0 = Bharat voices (Sarvam Bulbul). Agni Duplex = OpenAI GPT-Live 1, which we do not run. Agni 5.0 (Boson), 5.0 Lite (Svara), Lite 3.0 (Murf) and Lite 2.0 (Soniox) are vendors we do not run.

2. Official per-minute costs (vendor pricing pages, 24 Oct 2026, Rs96/$)

Item	$/min	Rs/min	Source
LiveKit agent session (every call)	0.0100	0.96	livekit.com/pricing
LiveKit observability (optional)	0.0100	0.96	livekit.com/pricing
Gemini 3.8 Live (audio in $3.00/M = $0.005/min; audio out $12/M = $0.018/min)	~0.0182 at LiveKit's blend	1.75	ai.google.dev/gemini-api/docs/pricing
OpenAI GPT-Live 1	0.0500	4.80	developers.openai.com/api/docs/pricing
OpenAI GPT Realtime 2.1 mini (audio $10/M in, $20/M out)	~0.0216	2.07	same
STT: AssemblyAI multilingual / Deepgram Nova-3 multilingual	0.0025 / 0.0058	0.24 / 0.56	livekit.com/pricing
TTS: Gemini 3.8 Flash-Lite TTS / Inworld 2.0 Flash (Scale)	0.0096 / 0.0054	0.92 / 0.52	livekit.com/pricing

Our measured per-minute (model cost only, LiveKit excluded): Flash-Lite + HD Rs0.90–1.15; Flash-Lite + Native Rs1.02; Nano + Native Rs0.53–0.76; Gemini Live 3.8 Rs4.7–5.3. Our LLM cost is ~8x LiveKit's estimate for the same model because every reply re-sends ~6–8K tokens, of which ~68% is our platform prompt and tool definitions (measured 23 Oct with agent/scripts/measure_prompt_tokens.py).

3. Function by function

Ravan column = [S] unless marked. Vistrow column = [V].

Area	Feature	Ravan	Vistrow	Verdict
Agent	Model choice	9 models in picker (see section 1)	Customers see 1 model (Vistrow Bharat); Live 3.8 and Nano Lab owner-only until combo tests finish (web-demo/src/lib/agentOptions.ts)	In progress on our side
Agent	LLM temperature	Slider in AI Provider Settings, default 0.7	None; "tone" preset changes the voice only (agent/main.py:349-383)	Behind
Agent	Voices	27 per Gemini model, 28 Soniox, 14 Boson, 8 Sarvam, 118 Murf	~157 customer voices (server/voice_catalog.py)	Ahead
Agent	Emotion	On/off on Gemini models and 5.0 (Boson); absent on Lite 2.0	`emotion_intensity` read at agent/main.py:4443 but no editor control (AgentDetail.tsx:197 defaults only)	Partial
Agent	Accent	Country + 21 Indian regional accents (Indian English; Hindi, Punjabi, Haryanvi, Bhojpuri, Pahadi; Rajasthani, Gujarati, Marathi, Konkani; Bengali, Odia, Assamese; Tamil, Telugu, Kannada, Malayalam; Manipuri, Mizo, Naga, Khasi)	None	Behind
Agent	Memory	Toggle; not on web calls; locked on Free; [A] not on Starter	`memory_enabled`, Starter and up; needs a caller phone number, so phone calls plus widget calls that pass `visitor_phone` (agent/main.py:4062-4064)	Equal
Agent	Prompt size	Word counter: 500 words on Free, 3,000 words on paid plan cards	No character limit. The editor shows a ~token counter and, on realtime models only, an amber/red warning above 1,000/2,000 tokens (AgentDetail.tsx:869-883). A pipeline-model warning is what was requested on issue #2	Ahead
Functions	End call	3/4/5 per plan	On by default; can be switched off through the agent's enabled-functions list (agent/main.py:3564-3571); no per-plan cap	Ahead
Functions	Transfer	[D] static or dynamic number, hold music, caller-ID option, transfer hours, GHL staff	Fixed `transfer_phone` only; colleague dialled into the same room; no hours, no summary (agent/tools.py:3657-3782)	Behind
Functions	Press digit	[D] agent presses digits on another menu, 1 s pause detection	`send_dtmf`, same idea (agent/main.py:3564)	Equal; neither has caller keypad menus
Functions	Custom function	[D] headers, query, JSON schema, save response to call variables, speak during/after	GET/POST/PUT/PATCH/DELETE, fixed headers, 10 s timeout, flat parameter list, first 2,000 chars to the model, no save-to-variables (agent/tools.py:3867-3923)	Behind
Speech	Transcription language	Auto-detect, Multilanguage, ~125 locales	17 Sarvam Indian codes pinned to the agent language; picker 10 Indian languages + English (agent/main.py:2353-2386, agentOptions.ts:472-484)	Equal for India
Speech	Background sound	None, City, Forest, Crowded, Keyboard, Hold	7 sounds + Off	Equal
Speech	Interruption sensitivity	Slider, default 0.9	Read at agent/main.py:7814; no editor control	Partial
Speech	Speech speed	Slider, default 1.0	None (pace fixed 1.0)	Behind
Speech	Reminder message	Seconds + times + custom text	`silence_reminder_ms` + max; no custom text (agent/main.py:7795-7813)	Equal minus text

Area	Feature	Ravan	Vistrow	Verdict
Call	Voicemail	Toggle	Outbound only, word pattern (English, Hindi, Telugu), fixed English line, not a setting (agent/main.py:1481-1490, 5561-5591)	Behind
Call	End on silence	Toggle + 10 s default	Setting read (agent/main.py:7819)	Equal
Call	Max duration	Slider, 30 min default	Up to 45 min, spoken wrap-up aimed at ~25 s before the cut (best effort) (agent/call_limit.py:15)	Ahead
Call	Emergency fallback	Toggle	`emergency_fallback_number` (agent/main.py:8596)	Equal
Call	Ring duration	Slider, 30 s default	Fixed 35 s (server/livekit_sip.py:189)	Behind
After call	Extraction	Custom fields, copy/paste sets, GPT-4o mini	`post_call_fields` + auto summary, sentiment, outcome, QA score (server/call_intelligence.py)	Equal, ours richer by default
After call	Webhook	Several URLs, custom header, retries; [D] unsigned	Account + per-agent URL; plain aiohttp POST, no URL safety check, no plan check, no signature (agent/tools.py:618-646); server integration deliveries do run the safety check (server/integrations_dispatch.py:48-59)	Behind + security bug on the agent path
CRM	CRM sync	Salesforce, GoHighLevel toggles	ArthaLeads, Zoho, Sheets, Slack, webhook	Different
Calendar	Booking	GHL or Cal.com per agent; own "Agni Calendar": event types, members, clients, branded email with domain auth, Google Calendar, Outlook, Calendly, iCloud (credentials only)	Built-in calendar: availability config (slot minutes), month/week/day/list views, statuses incl. rescheduled/cancelled, agent checks availability and books (agent/tools.py:1822, 2272); no external sync, no reschedule/cancel tool, no invites	Behind on sync
Knowledge	Normal KB	Text (multiple entries), URL add or Scan, files PDF/TXT/DOCX/MD, URL exclusions for auto-refresh, Refresh, Clear cache, Playground; searched as a tool during the call (`queryKnowledgeBase` seen in a transcript)	Text, URL (20 links), files, Q&A; whole text pasted into prompt, 8,000-char cap with a red warning bar (agent/db.py:513-557; KnowledgeBasePage.tsx:33,585); ~5,500 for realtime	Behind: no search, no refresh
Knowledge	Quick KB	Any file up to 500 MB incl. audio/video/scans (Gemini), links, text; Playground	None	Behind
Telephony	Providers	Twilio (own or managed), Telnyx (buy in-app, $1.20 US numbers, per-number concurrency + CPS), Indian (verify mobile → KYC with CIN/GSTIN docs → buy +91), WhatsApp calling, custom SIP (termination URI, CPS, user/pass)	EnableX only; Twilio/Exotel/Plivo "coming soon" (web-demo/src/pages/PhoneNumbers.tsx:51-54); numbers added one at a time; outbound trunks are per number (server/livekit_sip.py:119,145)	Behind: biggest gap
Inbound	Routes	Number → agent, timezone, max concurrent 1–5, credit budget cap, start/end dates, call window, limit calls per caller, active days	Number → agent, hours, days, start/end dates, per-number cap (agent/inbound_rules.py); outside hours = silent hang-up (agent/main.py:7270-7297)	Behind
Outbound	Campaigns	Wizard: name, agent, caller ID; step 2 needs a number; [D] list, schedule, Fresh/Follow-up/Failed-retry	Preflight, test dial, caller-ID rotation, retry policy per outcome (server/campaign_dialer.py, agent/retry_rules.py)	Ahead on controls
Calls	History + detail	Web/inbound/outbound filter, sentiment, model, credits, export; detail: credit breakdown, disconnect reason, inline tool calls, Report button	Filters search/channel/direction/feedback (client-side); recordings, transcripts, CSV (CallsHistory.tsx:97-193)	Equal

Area	Feature	Ravan	Vistrow	Verdict
Contacts	List + detail	CSV import, tags, status, variables, export; detail with Call button, stats, campaigns, activity, calls, notes	Same core + custom fields, notes	Equal
Team	Orgs and users	"+ New Organization" button but plan cards say 1 max organization ([A] confirms); roles User/Admin; email invites	One workspace per login (users.email UNIQUE, calls_db.py:202); roles viewer/member/admin/owner	Equal
Security	Login	Password, 2FA, sessions, Google sign-in	2FA with recovery codes; Google and GitHub sign-in (server/mfa.py)	Equal
Audit	Activity log	Org event log (logins with provider, KB changes)	Tenant security events; separate admin audit log; no tenant KB/agent change history	Behind
Plans in UI	Upgrade prompts	Locked Analytics page shows "isn't included in your plan … Upgrade"	Inbound, Outbound, Integrations have no plan wording; save fails with 403 (server/token_api.py:375-389)	Behind
Help	In-app assistant	"Ask Agni" chat; 2 of 7 answers wrong	Copilot that performs actions with confirmation, Growth+ (server/copilot_tools.py)	Ahead
Billing	Plans	Stripe checkout, credits, GST shown, invoice history	Razorpay, top-ups, GST invoices, annual	Equal

4. Ravan plan gates (their in-app plan cards [S])

Feature	Free	Starter Rs7,999	Growth Rs11,999	Scale Rs14,999
AI agents	0	3	4	5
Concurrent calls	5	3	5	5
Team members	0	5	5	10
Prompt length	500 words	3,000 words	3,000 words	3,000 words
End call / transfer / IVR functions	—	3 each	4 each	5 each
Custom functions	—	5	8	10
Post-call actions	—	10	10	10
Knowledge bases	—	2	2 sources	2 sources
Webhooks, voicemail detection	—	Yes	Yes	Yes
Call recording	Off	Included	Included	Included
Inbound / outbound campaigns	—	Not listed	10 + 10	15 + 15
Agent memory, reminder messages, background sound	—	Not listed	Yes	Yes
Calendar integration, analytics dashboard, custom phone provider, assign human agent	—	Not listed	Yes	Yes
"200 Kb Source Link Limit" ([A]: 200 KB of data across links)	—	—	200 KB	200 KB
API access	Off	Not included	Not included	Included

Contradictions in Ravan's own sources: the public pricing page ticks campaigns (5+5) and every product feature for Starter, while the in-app Starter card lists none of them; the assistant said Starter has no call recording (card: included) and has assign-human (card: Growth+); the FAQ says Starter is Hindi + English only; docs disagree on concurrency (8 / 15-25-50 / 5 / 1-10), model names and prices (Rs7,999 vs Rs8,000).

5. Vistrow blocks and bugs, ranked [V]

#	Problem	Type	Evidence	Fix
1	Outbound ceiling of 2 while plans sell 5/15/30	Block	Default 2 in server/campaign_dialer.py:90 and livekit_sip.py:194; read unset on Railway 24 Oct. Manual/test calls use this global ceiling; campaigns also cap by the trunk's and pool's configured lines (`_line_room`, campaign_dialer.py:164)	Buy EnableX channels and raise it, or sell 3/5/5
2	Website demo silent: Gemini prepay at –Rs23.37	Block	AI Studio billing; 402 errors in agent logs	Top up with auto-reload, or move agent 4 to Bharat
3	Scale's "premium voices" give customers none	Bug	Every premium entry (ElevenLabs and the Google preview voices) is owner-only for new selection (server/voice_catalog.py:107-108, 619-634; token_api.py `_guard_voice_tier`)	Open a premium set or drop the line
4	Voice cloning sold on Growth/Scale but off	Block	`VOICE_CLONE_OPEN` read unset on Railway 24 Oct (server/custom_voices.py); first clone would be free, later ones 25 credits	Open it or drop the line
5	Live-call webhook: no URL safety check, no plan check, no signature, and a 4xx/5xx reply is logged as "posted"	Security bug (agent path)	agent/tools.py:618-646 (status never read); server/integrations_dispatch.py:48-59 has the check	Sign, check URL and status, enforce plan
6	Customers get one model in the picker, but the API already accepts Live 3.8 from any tenant	Mismatch	agentOptions.ts MODEL_OPTIONS has one entry, ADMIN_ONLY_MODELS holds Nano Lab and Live 3.8; token_api.py `_TENANT_REALTIME_MODELS` lets tenants save Live 3.8	Owner decides: close the API until the combo tests finish, or open the picker
7	Telephony EnableX only, numbers one at a time	Gap	PhoneNumbers.tsx:51-54; livekit_sip.py:119,145	Generic SIP import, then Twilio
8	KB pasted into prompt, 8,000-char cap (warned), not searched; approved Q&A can itself exceed the cap	Gap	agent/db.py:513-557	Retrieval
9	Interruption sensitivity and emotion have no editor control	Bug	AgentDetail.tsx:197,205 defaults; read at agent/main.py:4443,7814	Add controls
10	No upgrade notice on Inbound/Outbound/Integrations; 403 on save	Bug	token_api.py:375-389	Upgrade screen
11	After-hours/over-cap inbound calls hang up silently	Gap	agent/main.py:7270-7297	Message or callback
12	No credit budget cap or per-caller limit on inbound routes	Gap	agent/inbound_rules.py	Add both
13	Transfer fixed number only; no hours; no summary	Gap	agent/tools.py:3657-3782	Hours, then dynamic
14	Voicemail message fixed English line	Gap	agent/main.py:5561-5591	Make it a setting
15	No temperature/speech-speed; ring fixed 35 s	Gap	agent/main.py:349-383; livekit_sip.py:189	Add with model picker
16	No automatic brochure sending (flags CRM outcome for sales)	By design	agent/tools.py:2157-2177	Optional WhatsApp auto-send later
17	Old orchestrator pipeline's transfer is a stub; new accounts no longer use it (19 Oct)	Low	orchestrator/tools.py:577-599; calls_db.py:1626-1634	Remove or fix before reuse

#	Problem	Type	Evidence	Fix
18	Custom functions call any URL with no public-URL check, at save or at call time	Security bug	agent/tools.py:3910-3918 (plain aiohttp); no custom-function URL check in server/token_api.py	Same `_assert_public_url` + safe opener as integrations
19	KB cache keyed by KB id only, but stores text already cut to the caller's limit, so realtime (~5,500) and pipeline (8,000) can get each other's length	Bug	agent/db.py:534 (get), 557 (cut), 567 (`_kb_cache[kb_id]`)	Key by (kb_id, max_chars)
20	Chirp 3 STT runs in the US region by default (Asia regions 403 for hi-IN)	Latency	agent/main.py:2491-2493	Measure before promising Asia latency
21	Concurrency sold up to 30, LiveKit Ship lists 20 concurrent agent sessions	Unverified	livekit.com/pricing (Codex); our subscription limits not checked	Check the LiveKit project's limit

6. Corrections made during the audit (earlier drafts were wrong)

- Ravan's "IVR / Press Digit" is agent-side (presses digits on another menu), same as our `send_dtmf`; not a gap.
- Ravan's prompt cap is 3,000 words (500 on Free), about 18,000 characters — not 3,000 characters.
- Our follow-up "days after the call" setting works (steps editor sets it on client and server, retry_rules.py:208-211); removed from the bug list.
- New signups no longer default to the orchestrator pipeline (fixed 19 Oct, calls_db.py:2096-2103).
- Our KB cap is not silent; the KB page shows a red warning bar.
- Our brochure tool is by design, not a bug.
- Our server-side integration deliveries do run the URL safety check; only the agent path lacks it.
- Ravan's calendar syncs Google, Outlook and Calendly, not only GHL/Cal.com. ([S] on the Agni Calendar screen; Codex could not reload it, so treat as unconfirmed.)
- Ravan paid plans allow 1 organization; "200 Kb Source Link Limit" is 200 KB of data. ([A] only: unconfirmed.)
- Ravan's models map to third-party vendors by catalog alias; in-house processing is not ruled out (see section 1).

7. Codex cross-check, reconciled (24 Oct 2026)

Codex reviewed this audit against `origin/main` db9d1e17 (newer than 9ef663cf). Each disputed point was rechecked by Claude before changing the text above. "Rechecked" = Claude read the code at db9d1e17, re-read Ravan's catalog, or read the vendor's own page. "Codex only" = accepted as Codex's finding but not rechecked by Claude.

Point	Result	How it was checked
Ravan catalog: 11 entries, 9 visible, 3 with empty `sessionMode`	Original wrong (said 8 visible, all `pipeline`)	Rechecked: same read-only GET of the catalog, 24 Oct
Emotion is Gemini-only	Original wrong (own [S] note showed 5.0 keeps it)	Rechecked: catalog says 5.0 = `boson`
"Ravan has no in-house models"	Overstated; narrowed	Catalog shows aliases only
Bharat short names are Bulbul v2	Original wrong: v4 Flash	Rechecked agent/main.py ~3290
`google31:` is Gemini 2.5 TTS on Vertex	Original wrong: `gemini-3.1-flash-tts-preview`. `google:kore/charon` are the 2.5 voices	Rechecked agent/main.py `_GOOGLE_31_MODEL`, voice_catalog `GEMINI_TTS_MODELS`
Kore/Charon open to all	Original wrong: premium + preview, owner-only	Rechecked voice_catalog.py:107-108, token_api `_guard_voice_tier`
Live 3.8 owner-only	Half right: picker yes, API no; billed 2 credits/min	Rechecked agentOptions.ts:323-332, token_api `_TENANT_REALTIME_MODELS`, agent/plan_policy.py
Memory phone-only	Original wrong: any call with a caller phone, incl. widget `visitor_phone`	Rechecked agent/main.py:4062-4064
End call always on	Original wrong: default on, can be disabled	Rechecked agent/main.py:3564-3571
Outbound 2 calls everywhere	Qualified: global ceiling for manual/test; campaigns also by trunk/pool lines	Rechecked campaign_dialer.py:86-90, 164; livekit_sip.py:194
Every clone costs 25 credits	Original wrong: first clone free on Growth/Scale	Rechecked custom_voices.py:26-30, 70-73
Prompt has no warning	Original wrong: realtime-only token warning exists	Rechecked AgentDetail.tsx:855-883

Point	Result	How it was checked
Webhook status not checked; custom functions no URL check; KB cache key	New bugs 5, 18, 19	Rechecked agent/tools.py:630-646, 3910-3918; agent/db.py:534-567; no URL check in token_api.py
Chirp STT US region	New item 20	Rechecked agent/main.py:2491-2493
Murf Falcon price unit	Original wrong: $0.01 per 1,000 characters (marketing says per minute)	Rechecked murf.ai Falcon docs page
Soniox sells TTS	Correct; Lite 2.0 can be Soniox end to end	Already in the vendor plan; Codex's voice-name match is Codex only
Svara = Kenpath	Unproven	Codex only (`sv_` id convention)
GPT-Live voices, paid-tier access, Hindi voice `nira`	Codex only	Not rechecked
Boson TTS Indian languages; LiveKit integration "coming soon"; open weights non-commercial	Codex only (licence point matches the plan)	Not rechecked
LiveKit Ship 20 concurrent sessions	Codex only; item 21	Not rechecked
Gemini balance –Rs23.37 / 402s; Railway variables unset	Claude saw these on 24 Oct (AI Studio billing, agent logs, Railway variables); Codex could not	Production state; recheck before acting
Measured Rs/min, 8x, 68% prompt share	Claude's measurements (tmp/combo-test, agent/scripts/measure_prompt_tokens.py); not independently reproduced	Rs96/$ is an assumed rate

Still open (not verified by anyone): Ravan's exact backend model versions and stage combinations, Ravan paid-plan entitlements beyond the cards, comparable latency/quality/cost on equal test calls.